



Garnet

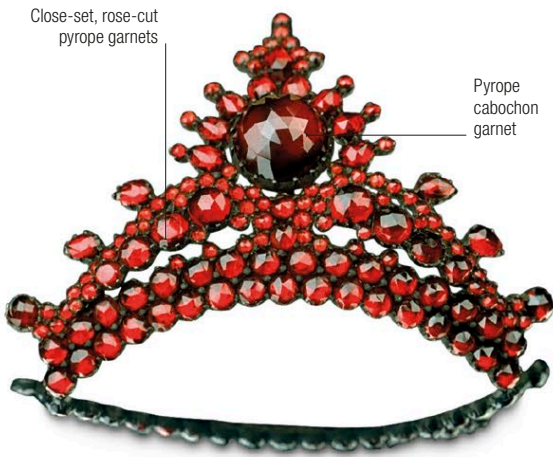
△ Round, brilliant-cut almandine garnet

Garnets are generally thought of as red, but they can also be orange, pink, green, black, and honey-brown. All species of garnets have similar physical properties and crystal forms, but differ in chemical composition. There are over 15 garnet species, of which six varieties are most commonly used as gems: pyrope, almandite, spessartite, grossularite (includes hessonite and tsavorite), andradite (includes demantoid), and uvarovite. Although they are found in many different colours and compositions, garnets are easily recognized because they are generally found as well-developed crystals with a basic – although sometimes modified – dodecahedral form. The name “garnet” is derived from the Latin *granatus*, from granum (“grain, seed”), possibly a reference to the vivid red seed covers of the pomegranate, which are similar in shape, size, and colour to some garnet crystals.

A tradition of inlaying

The use of garnets as gemstones dates back to at least the Bronze Age. They were especially used inlaid in gold cells in the cloisonné technique, a style often just called garnet cloisonné. Many consider this the highest point of garnet work, and it can be seen in Anglo-Saxon England at Sutton Hoo and in the Staffordshire Hoard (see pp.264–65). A lesser known use of garnet is as an abrasive, used instead of silica sand in sand blasting. Garnet is used to cut steel and other materials in high-pressure water jets. Garnet sandpaper is a favourite of cabinet makers for finishing bare wood.

Key pieces



Antique hairpin | Set with pyrope garnets from Bohemia (now part of the Czech Republic), this gold hairpin was probably crafted in Victorian times. Pyrope, from the Greek *pyropos*, means “firelike”.



Stork pendant | Set with a large and luxurious garnet cabochon and highlighted with baroque pearls, this gold pendant with facing storks dates from about 1900. Storks often symbolize purity and renewal.



Fabergé seahorse brooch | This seahorse wrapped in seaweed is set with green demantoid garnets, tsavorite garnets, alexandrites, tourmalines, sapphires, and diamonds.

Specification

Chemical name (A)Ca/Fe/Mg/Mn (B)Al/Cr/Si/Ti/Zr/Vn Silicate
Formula $A_3B_2(SiO_4)_3$ | **Colours** Black, brown, yellow, green, red, violet, orange, pink | **Structure** Cubic | **Hardness** 7–7.5
SG 3.6–4.3 | **RI** 1.73–1.94 | **Lustre** Vitreous | **Streak** White



Emerald



Step



Round brilliant



Oval brilliant



Cabochon



Locations
1 Canada 2 USA 3 Mexico 4 Germany 5 Czech Republic
6 Italy 7 Namibia 8 South Africa 9 Kenya 10 Tanzania
11 Madagascar 12 Sri Lanka